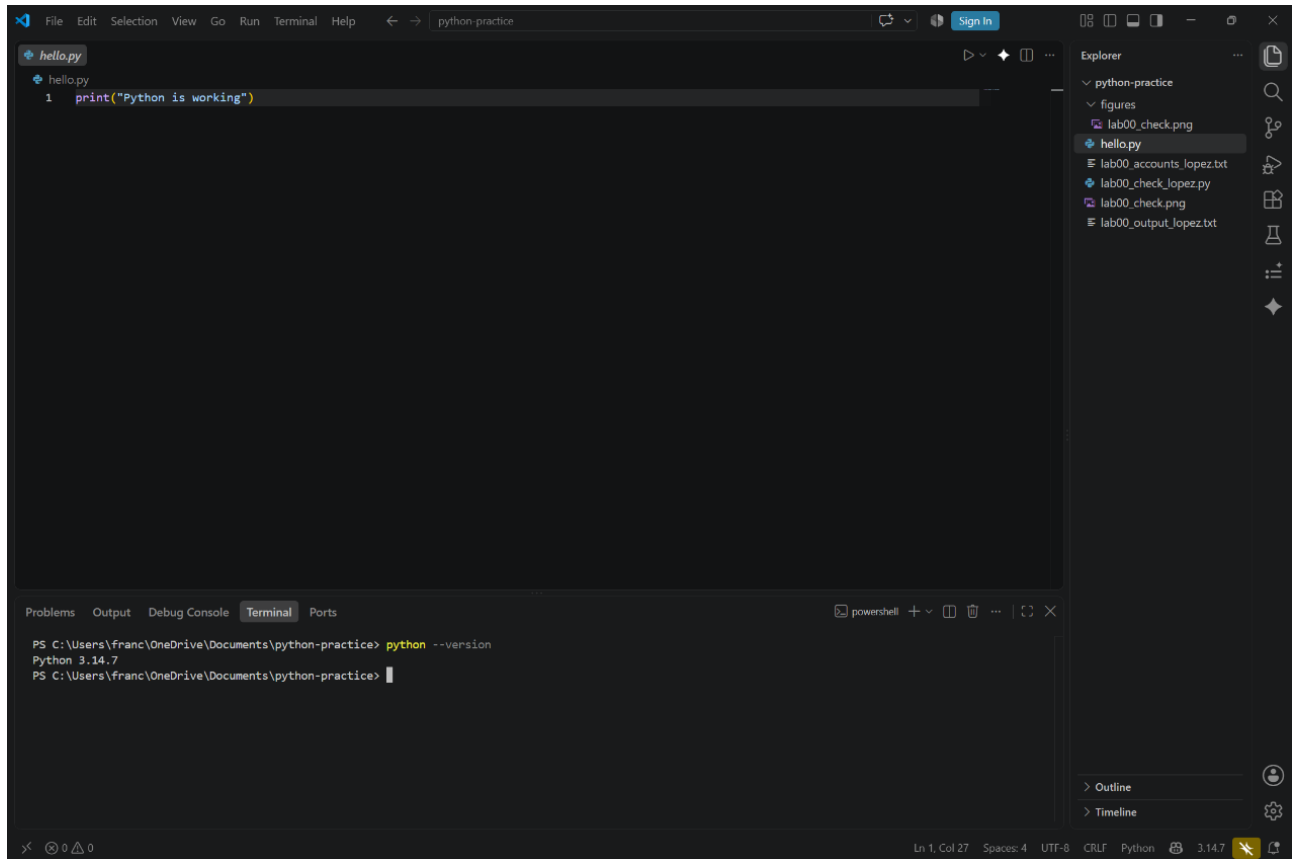
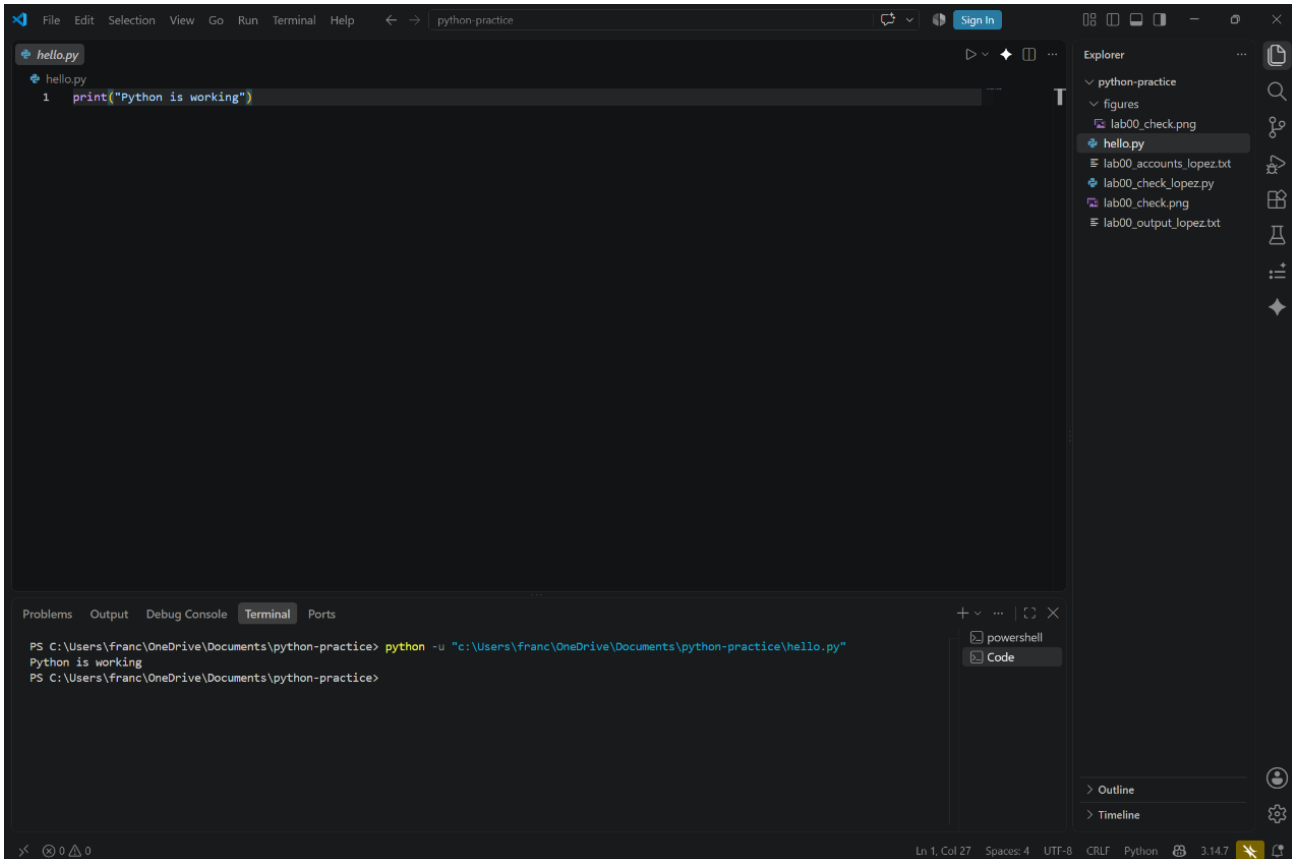


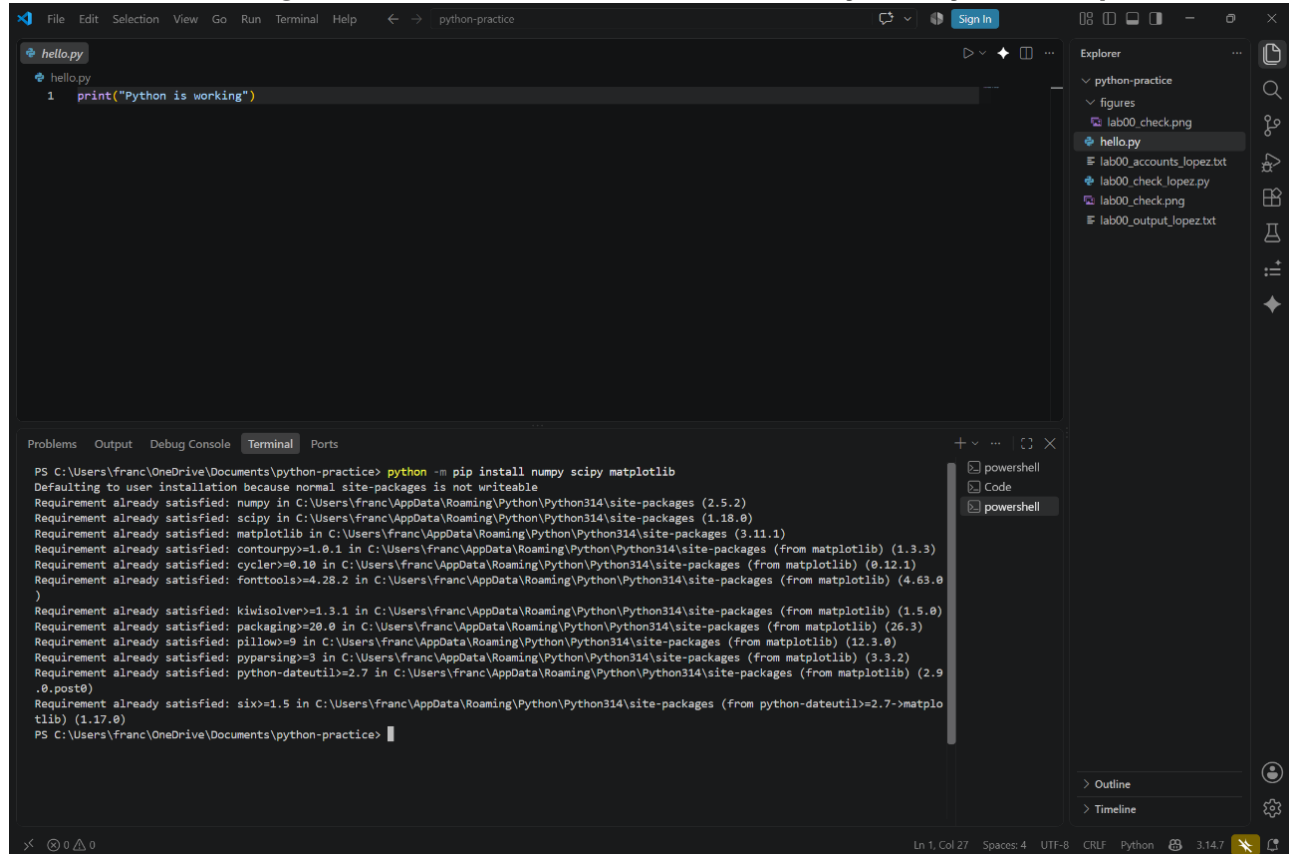
1. The terminal showing python --version and its answer.



2. VS Code with hello.py open and its output visible in the terminal.



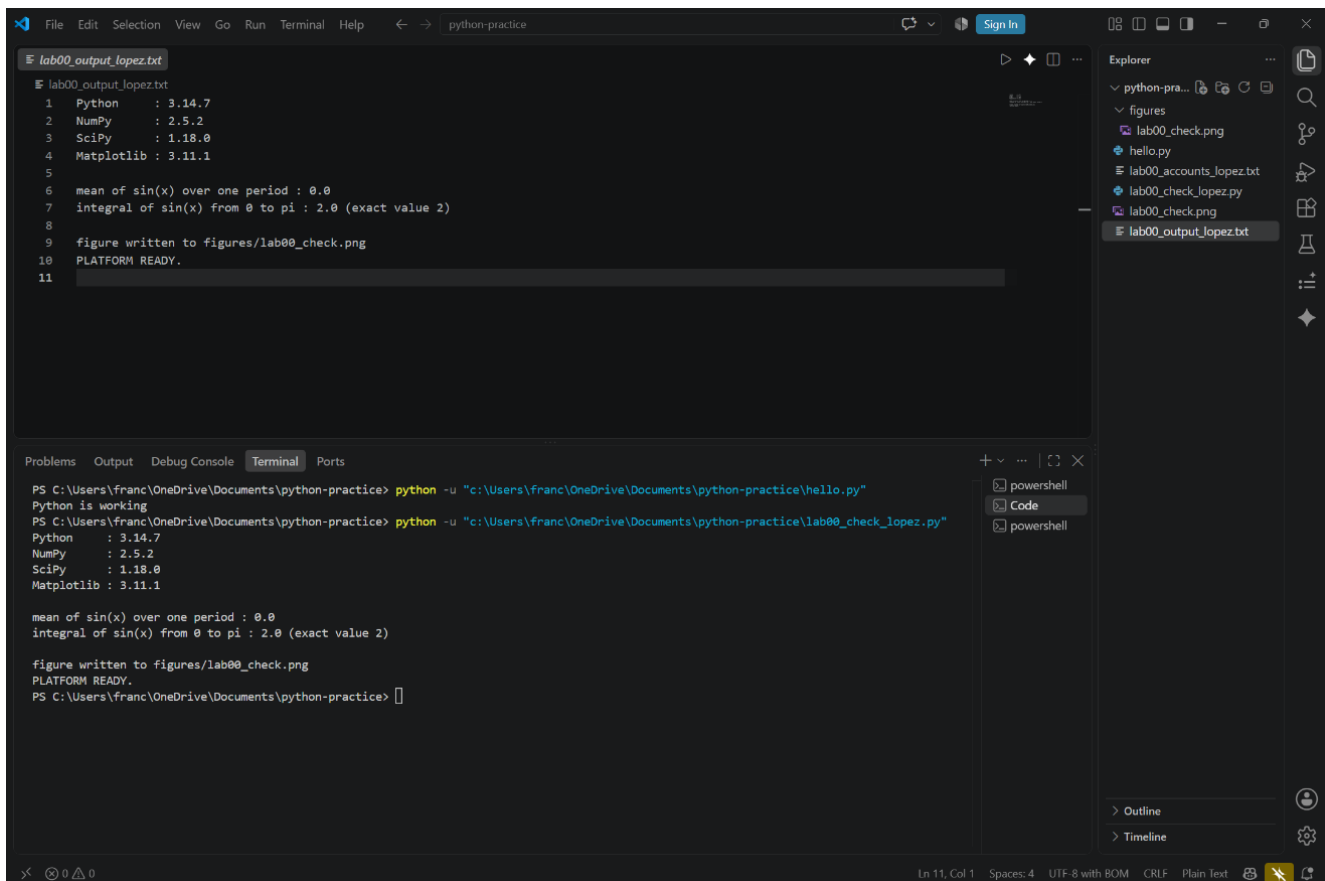
3. Terminal showing the successful installation of NumPy, SciPy and Matplotlib.



The screenshot shows the Visual Studio Code interface with a file named `hello.py` open in the editor. The file contains a single line of Python code: `print("Python is working")`. The Explorer sidebar on the right shows the file structure, including `hello.py` and several other files. The Terminal panel at the bottom shows the output of the command `python -m pip install numpy scipy matplotlib`. The output indicates that all requirements are already satisfied, confirming the successful installation of the packages. The status bar at the bottom shows the current line and column (Ln 1, Col 27) and the file encoding (UTF-8).

```
PS C:\Users\franc\OneDrive\Documents\python-practice> python -m pip install numpy scipy matplotlib
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (2.5.2)
Requirement already satisfied: scipy in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (1.18.0)
Requirement already satisfied: matplotlib in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\franc\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\franc\OneDrive\Documents\python-practice>
```

4. Full console output of verification program, ending in PLATFORM READY.



The screenshot shows the Visual Studio Code interface with a file named `lab00_output_lopez.txt` open in the editor. The file contains the output of a verification program, including the Python version, NumPy, SciPy, and Matplotlib versions, and the results of a calculation. The Terminal panel at the bottom shows the command `python -u "c:\Users\franc\OneDrive\Documents\python-practice\hello.py"` and the output of the program. The status bar at the bottom shows the current line and column (Ln 11, Col 1) and the file encoding (UTF-8 with BOM).

```
Python      : 3.14.7
NumPy       : 2.5.2
SciPy       : 1.18.0
Matplotlib  : 3.11.1

mean of sin(x) over one period : 0.0
integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY.
```

```
PS C:\Users\franc\OneDrive\Documents\python-practice> python -u "c:\Users\franc\OneDrive\Documents\python-practice\hello.py"
Python is working
PS C:\Users\franc\OneDrive\Documents\python-practice> python -u "c:\Users\franc\OneDrive\Documents\python-practice\lab00_check_lopez.py"
Python      : 3.14.7
NumPy       : 2.5.2
SciPy       : 1.18.0
Matplotlib  : 3.11.1

mean of sin(x) over one period : 0.0
integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY.
PS C:\Users\franc\OneDrive\Documents\python-practice>
```

5. My GitHub repository page

The screenshot shows the GitHub repository page for 'numerical-methods-lab' by francinejoy-lopez. The repository is private and has 0 Watchers, 0 Forks, and 0 Stars. The main branch is 'main' with 1 Branch and 0 Tags. The repository contains two files: 'README.md' (Initial commit, 25 minutes ago) and 'hello.py' (Add files via upload, 24 minutes ago). The README file is displayed, showing the title 'numerical-methods-lab'. The right sidebar shows the 'About' section with no description, website, or topics provided. It also lists 'Releases' (No releases published), 'Packages' (No packages published), 'Contributors' (1 contributor: francinejoy-lopez), and 'Languages' (Python 100%).

The Figure

